	Benha University - Faculty of Computer and Artificial Intelligence		
	1st Term (NOV 2025) – Midterm Exam		
	Program: General – Second Year Students		
	Course Name: Logic Design	Course Code: CS 221	Level: 2nd level
	Date: 8 / 11 / 2025	Time: 45 minutes	Exam in 2 Pages
	Total Marks: 15 Marks	Credit : 3 Hours	

Student Name : ..... Section: ..... Student Number .....

Question 1: Answer the following [ILO's: A1, A4,A11,B1,B5,B6,D2] [6 Marks]

1-1 Convert the following numbers and show your workout. (2 marks)

a) $(261012)_8$ into binary number

(0101100010000001010)2

b) $(10110001000001010)_2$ into hexadecimal number

0001 0110 0010 0000 1010

1 6 2 0 A

1-2 Perform the following 4-bit binary numbers operation and give the results in the decimal system (base 10). (2 marks)

a) 1100 – 0110 (use the 2's complement notation)

two's comp of 0110 invert 1001 + 1 1010

1100+1010 = 10110 ignore carry 110

b) 0111 + 1001

0111+1001=10000

1-3 Convert the following from binary to Gray code and vice versa. (2 marks)

a) $(11001)_2$

10101

b) (10011) gray code

11101

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Question 2 : Answer the following [ILOs: A2,A3,A11,B1,B5,B6,D3] [9 Marks]

2-1 For the given function in Table:

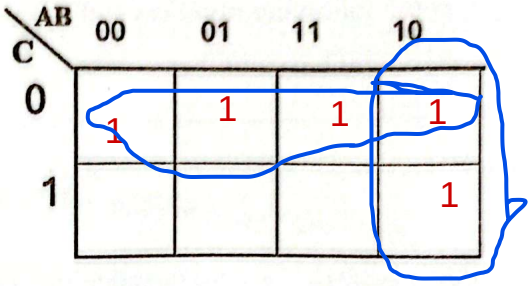
ABC	F
0 0 0	1
0 0 1	0
0 1 0	1
0 1 1	0
1 0 0	1
1 0 1	1
1 1 0	1
1 1 1	0

a. (2 mark) Find the SOP expression for the function F

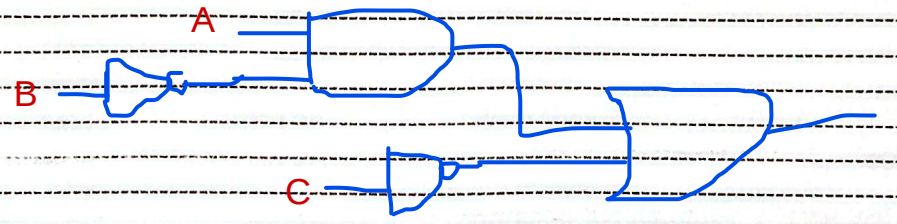
F=A'B'C'+A'BC'+AB'C'+AB'C+ABC'

b. (2 marks) Find the most simplified SOP form using K-Map

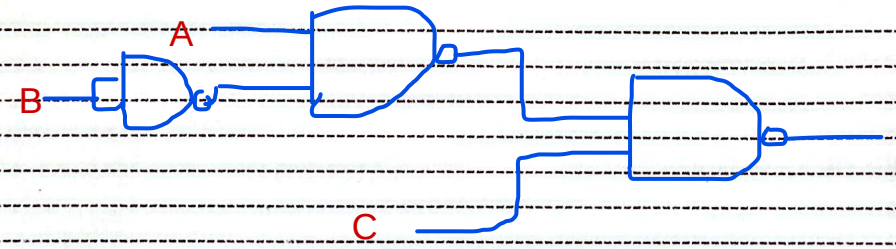
F=C'+AB'



c. (2 marks) Draw the circuit diagram for the SOP using only 2-input AND-OR gates.



d. (2 marks) Draw the circuit diagram for the SOP using only 2-input NAND gates.



e. (1 mark) Derive a standard POS expression.

F = F=(A+B+C')(A+B'+C')(A'+B'+C')

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Dr. Fatma Sakr

Dr. Doaa Al Ebiary

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Question 1: Answer the following [ILOs: A1,A4,A11,B1,B5,B6,D2] [6 Marks]

1-1 Convert the following numbers and show your workout. (2 marks)

a) $(B10A)_{16}$ into binary number

(1011000100001010)2

b) $(1011000100001010)_2$ into octal number

(261012)8

1-2 Perform the following 4-bit binary numbers operation and give the results in the decimal system (base 10). (2 marks)

a) 1100 – 0110 (use the 2's complement notation)

two's comp of 0110 invert 1001 + 1 1010
1100+1010 = 10110 ignore carry 110

b) 0111 + 1001

0111+1001=10000

1-3 **Convert** the following from binary to Gray code and vice versa. (2 marks)

a) $(11001)_2$

10101

b) (10011) gray code

11101

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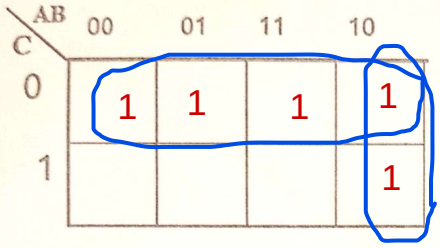
Question 2 : Answer the following [ILOs: A2,A3,A11,B1,B5,B6,D3] [9 Marks]

2-1 For the SOP expression $F = \bar{A}\bar{C} + A\bar{B} + AB\bar{C}$:

a. (2 marks) List the Truth Table of F

ABC	F
0 0 0	1
0 0 1	0
0 1 0	1
0 1 1	0
1 0 0	1
1 0 1	1
1 1 0	1
1 1 1	0

b. (2 marks) Find the most simplified SOP form using K-Map



F=C'+AB'

c. (2 marks) Draw the circuit diagram for the SOP using only 2-input AND-OR gates.

d. (2 marks) Draw the circuit diagram for the SOP using only 2-input NAND gates.

e. (1 mark) Derive a standard POS expression.

F = F=(A+B+C')(A+B'+C')(A'+B'+C')

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